EXCHANGE OF RESPIRATORY GASES

Dr. Fatima Haq

Lecturer

DIKIOHS, DUHS.

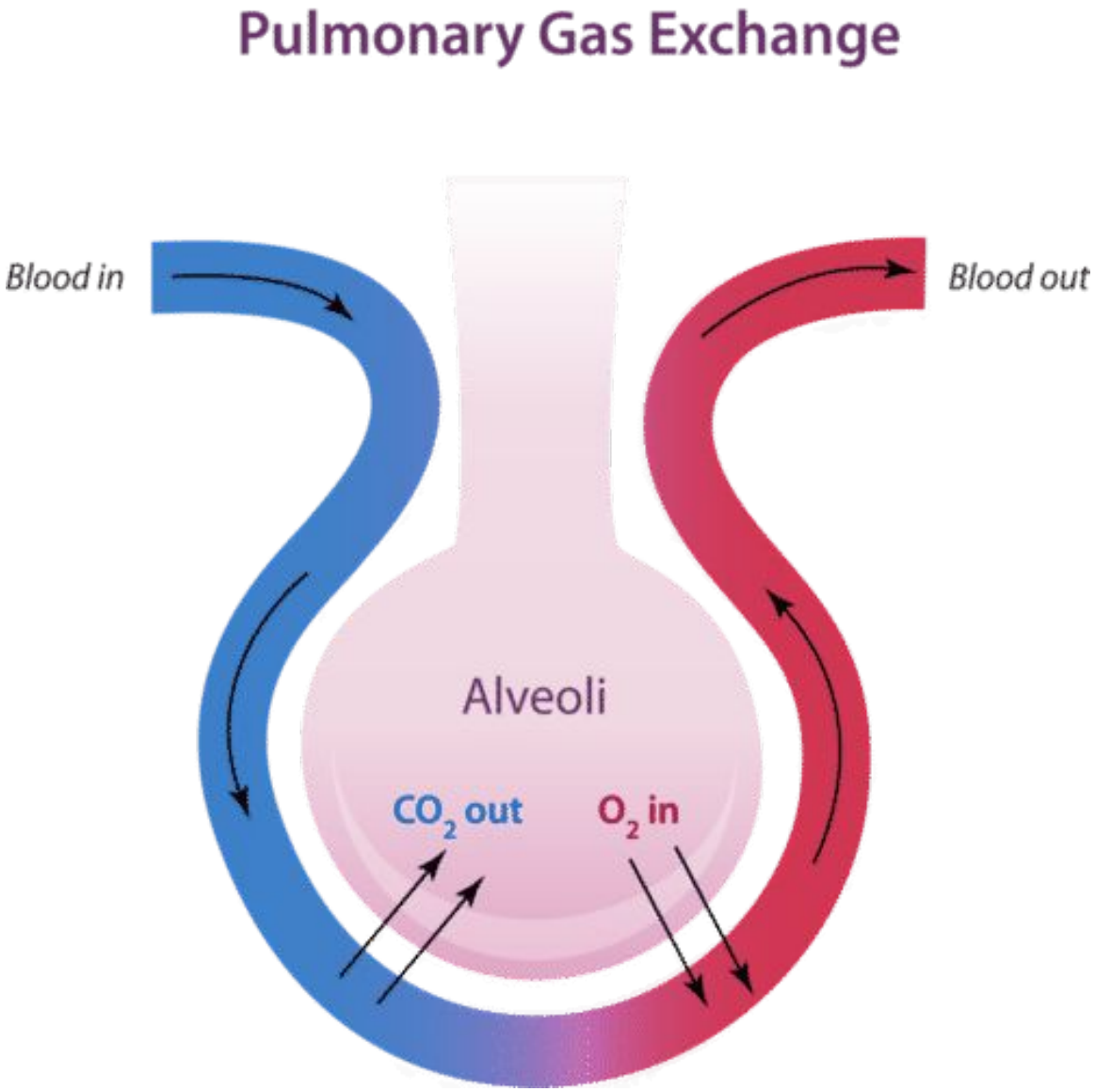
LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- ✔ Explain the mechanism of exchange of oxygen & carbon dioxide across the respiratory membrane.
- ✔ Describe the process of diffusion of respiratory gases between alveoli, blood & tissues.
- ✔ Identify the factors affecting gas exchange including partial pressure gradients, surface area, membrane thickness & gas solubility.
- ✔ Define ventilation-perfusion (V/Q) relationship & normal V/Q ratio.
- ✔ Identify variations of V/Q ratio in different regions of the lungs.
- ✔ Explain the effects of V/Q mismatch on oxygenation & carbon dioxide elimination.

INTRODUCTION

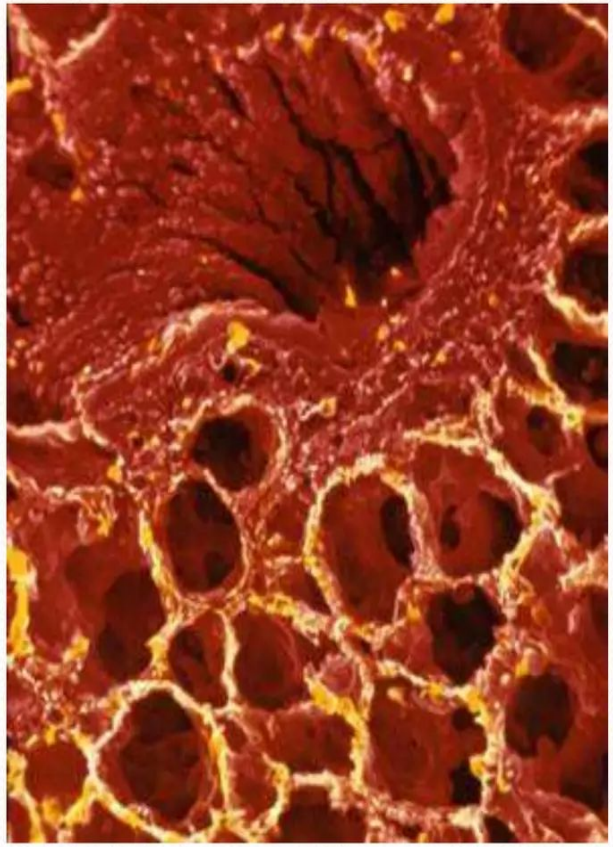
- After the ventilation of alveoli with fresh air, the next step in respiration is the exchange of oxygen from the alveoli into the pulmonary blood & carbon dioxide in opposite direction(out of blood into the alveoli).
- This exchange of respiratory gases in the lungs takes place through ***bulk flow diffusion*** across the respiratory membrane.



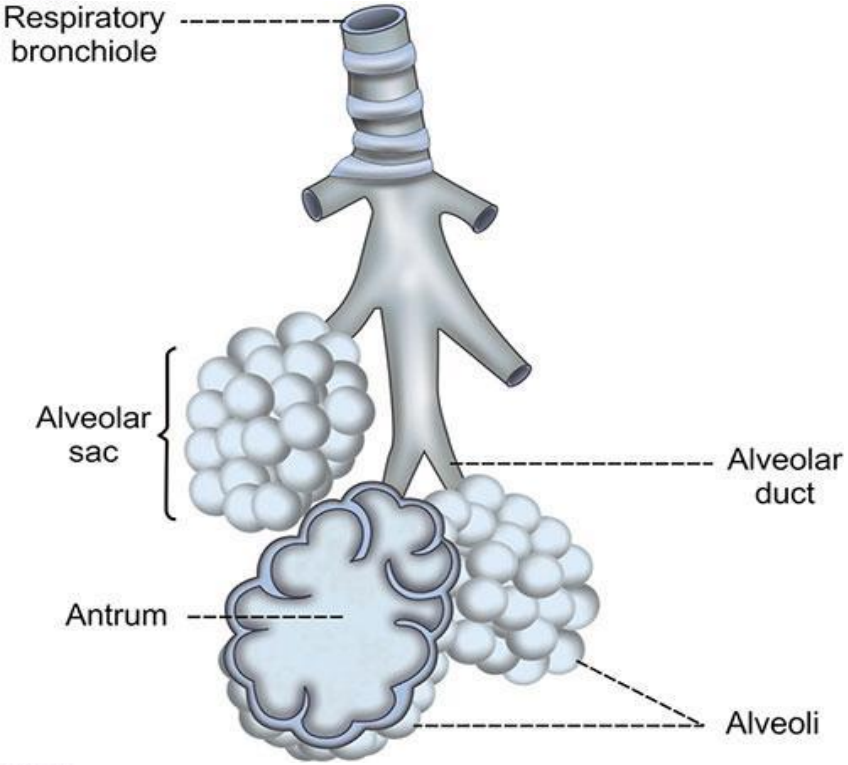
- Respiratory Unit

Respiratory unit is the terminal portion of the Respiratory Tract. It includes:

- 1. Respiratory Bronchioles
- 2. Alveolar Ducts
- 3. Antrum
- 4. Alveolar Sacs
- 5. Alveoli (Human beings have a thin layer of about 700 million alveoli within their lungs. Which is crucial for respiration, exchanging O_2 & CO_2 with the surrounding blood capillaries.)



Microscopic Structure of Alveoli



RESPIRATORY MEMBRANE

It is a membranous structure through which exchange of respiratory gases occur.

- Formed by the *epithelium* of respiratory unit & *endothelium* of pulmonary capillary.
- Total surface area of respiratory membrane in the lungs is about 70 sq.m.

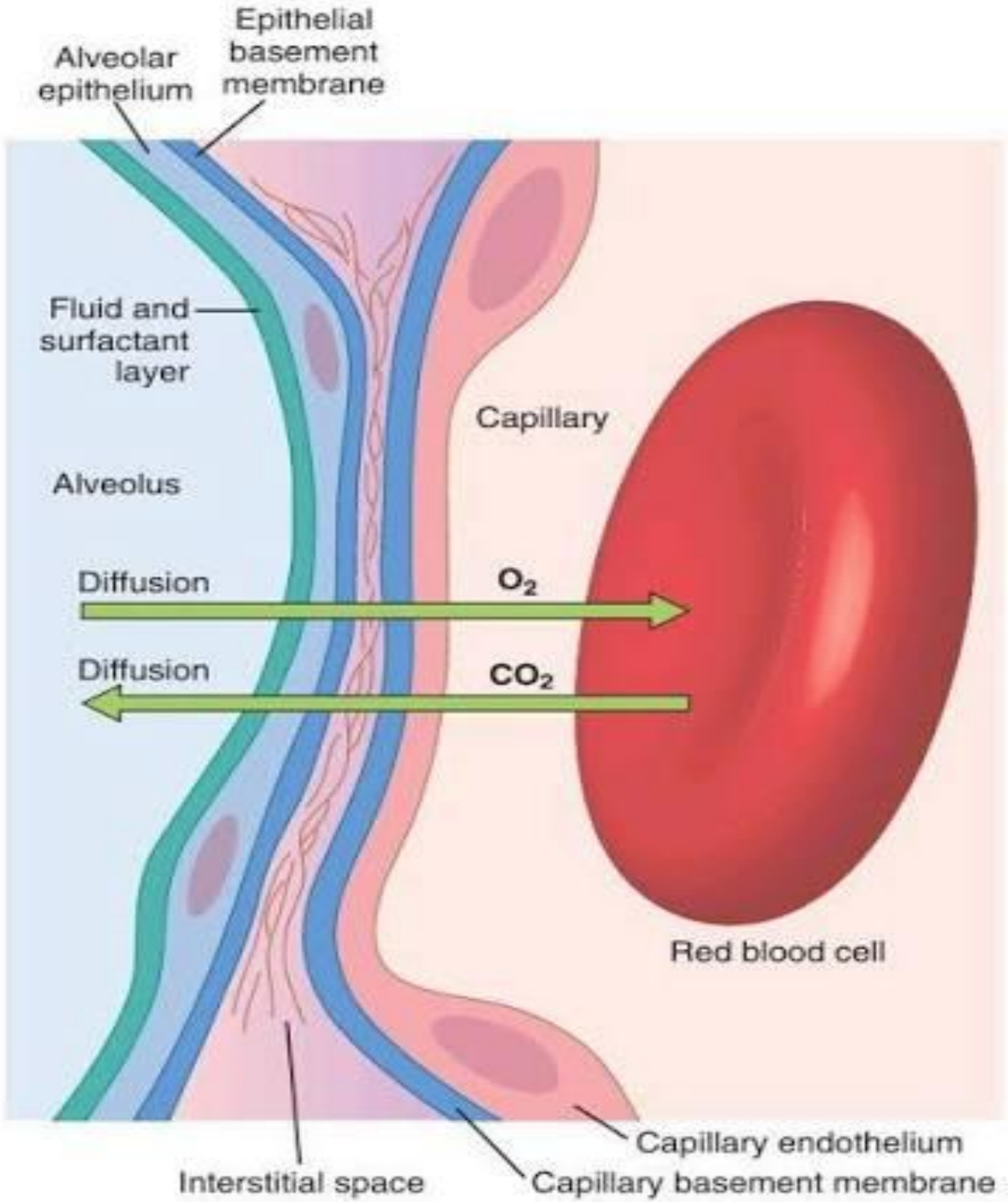
LAYERS OF RESPIRATORY MEMBRANE

- Respiratory membrane is formed by different layers of structures of alveoli & capillaries i.e.

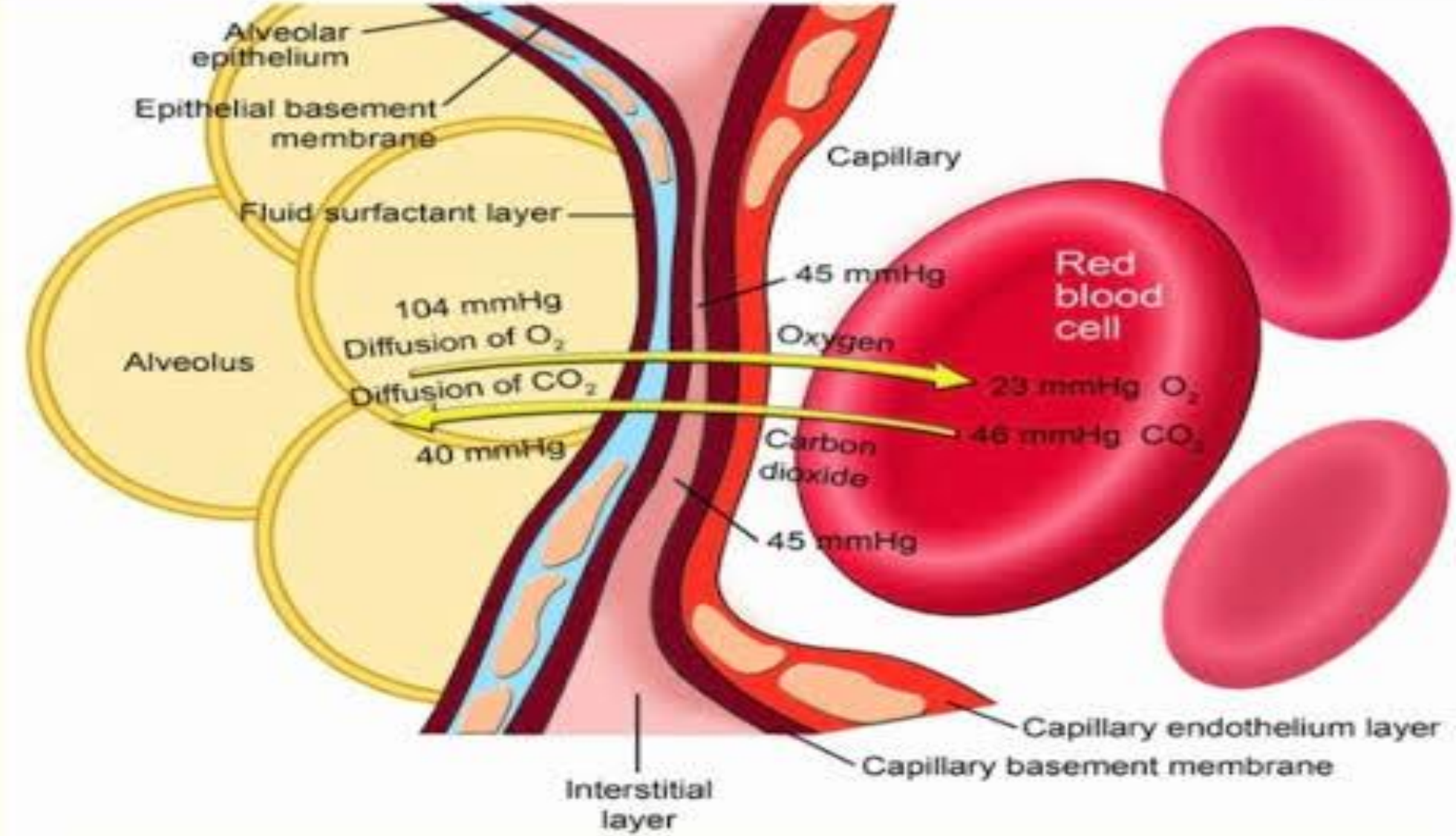
1. Layer of fluid surfactant that lines the alveolus
2. Alveolar epithelium
3. Epithelial basement membrane
4. Interstitial space between alveolar epithelium & capillary membrane
5. Capillary basement membrane(fuses in many places with alveolar epithelial basement membrane)
6. Capillary endothelium

Cont..

- Despite having many layers, it is very thin with an average thickness of ***0.6 micron***.
- Membrane of RBCs is in close contact with the capillary wall as average diameter of pulmonary capillary is 8 micron whereas average diameter of RBC is 7.4 micron which facilitates quick exchange of O_2 & CO_2 between blood & alveoli.



LAYERS OF THE RESPIRATORY MEMBRANE



Henry's Law

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DEFINITION

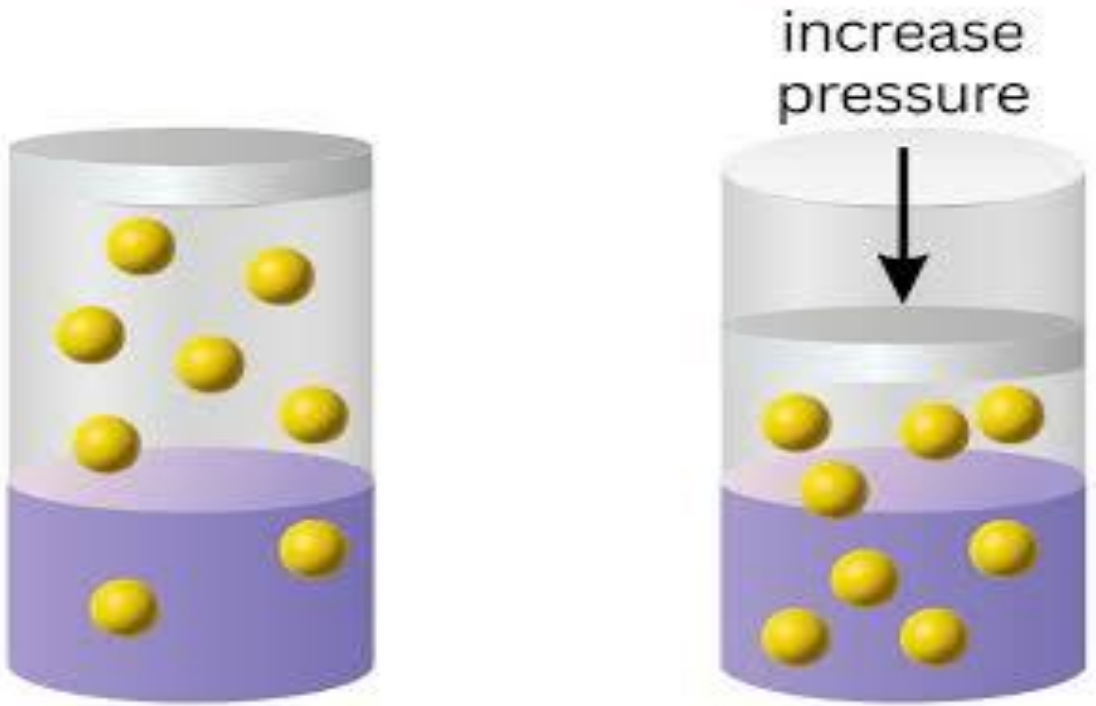
At a constant temperature, the amount of a gas dissolved in a liquid is directly proportional to the partial pressure of that gas above the liquid.

FORMULA

$$C = k_H P$$

- C = concentration of dissolved gas (mol/L)
- k_H = Henry's law constant ($mol \cdot L^{-1} \cdot atm^{-1}$)
- P = partial pressure of the gas (atm)

Illustration: Effect of Pressure on Gas Solubility



low pressure → fewer molecules enter the liquid, more escape

high pressure → more molecules enter the liquid, fewer escape

As the partial pressure (P) of the gas increases, the concentration (C) of the dissolved gas increases proportionally.

DIFFUSING CAPACITY

- It can be defined as:

The volume of gas that diffuses through the respiratory membrane each minute for a pressure gradient of 1mm Hg.

$$DC \propto \frac{Pg \times S \times A}{Mw \times D}$$

DC = Diffusing capacity
Pg = Pressure gradient
S = Solubility of gas
A = Surface area of respiratory membrane
Mw = Molecular weight
D = Thickness of respiratory membrane.

- Diffusing capacity for O_2 is 21 mL/min/1mm Hg whereas for CO_2 it is 400mL/min/1mmHg.
- So, the diffusing capacity of respiratory membrane for CO_2 is about 20 times greater than that for O_2.

FACTORS AFFECTING DIFFUSING CAPACITY

1. PRESSURE GRADIENT

- Pressure gradient is:

  the difference between the partial pressure of a gas in alveoli & pulmonary capillary blood.

- Diffusing capacity is directly proportional to pressure gradient.

2. SOLUBILITY OF GAS IN FLUID MEDIUM

- Diffusing capacity is directly proportional to solubility of a gas i.e. greater the solubility of a gas in the fluid medium means greater no. of molecules dissolve in it & diffuse easily.

3. TOTAL SURFACE AREA OF RESPIRATORY MEMBRANE

- Diffusing capacity is directly proportional to surface area of respiratory membrane.
- Surface area of respiratory membrane in each lung is about 70 sq.m.
- In ***emphysema***, diffusing capacity is decreased as many alveoli are collapsed due to heavy smoking or oxidant gases.

4. MOLECULAR WEIGHT OF THE GAS

- Diffusing capacity is inversely proportional to the molecular weight of the gas.
- More is the molecular weight of a gas more is the density & less is the rate of diffusion.

5. THICKNESS OF RESPIRATORY MEMBRANE

- Diffusing capacity is inversely proportional to the thickness of respiratory membrane.
- In conditions like ***fibrosis*** & ***edema***, increased thickness of respiratory membrane causes decreased diffusion rate because the distance through which the diffusion occur becomes longer.

DIFFUSION COEFFICIENT & FICK'S LAW OF DIFFUSION

What is diffusion coefficient?

- It is defined as:

A constant(factor of proportionality) which is the measure of a substance diffusing through the concentration gradient.

- It is also known as ***diffusion constant*** & is related to the size & shape of molecules of the substance.

FICK'S LAW OF DIFFUSION

- This law states that:

Amount of a substance crossing a given area is directly proportional to the area available for diffusion, concentration gradient & a constant called diffusion coefficient.

So,

Amount diffused = area x concentration gradient x diffusion coefficient.

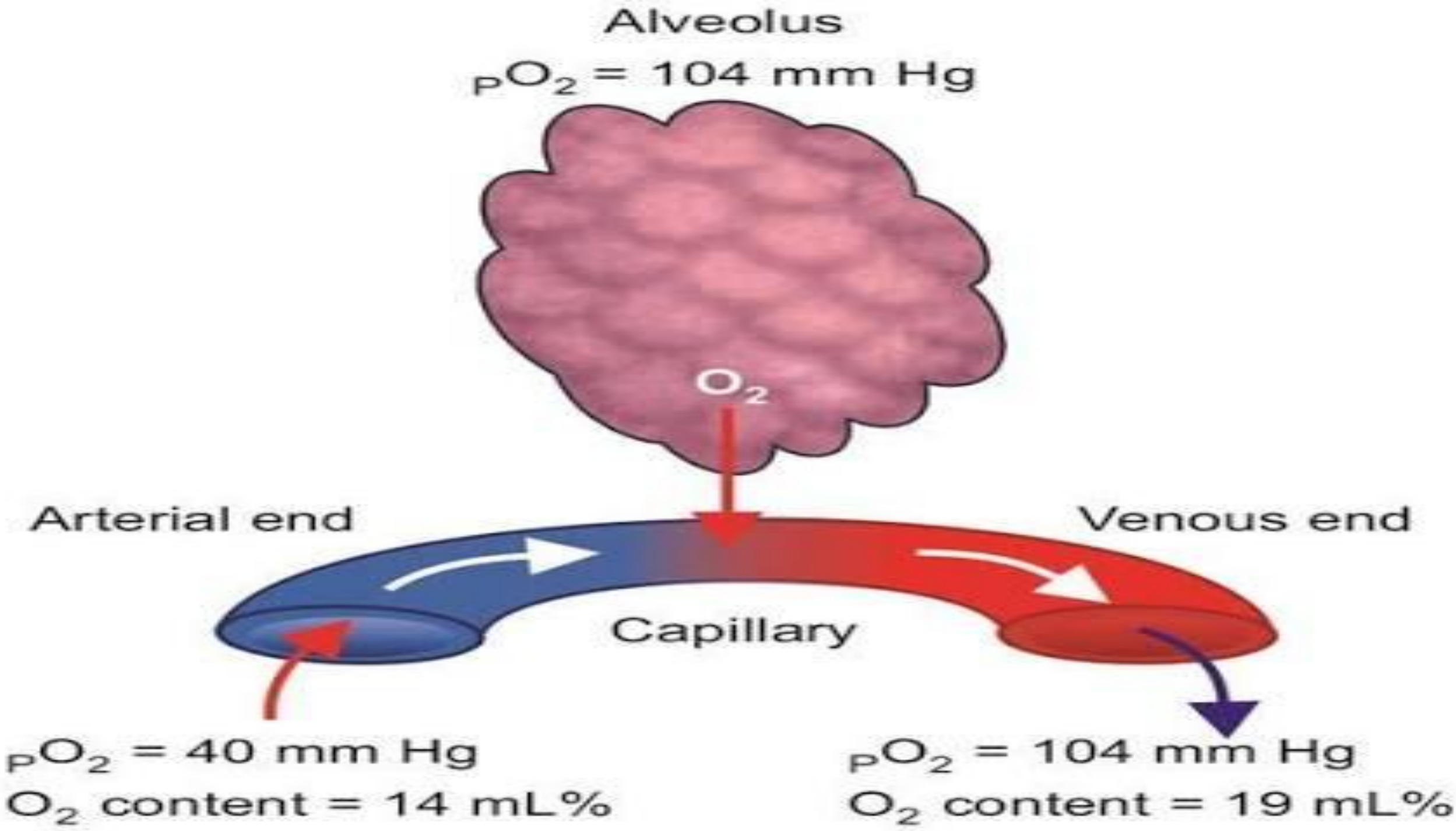
DIFFUSION OF OXYGEN

FROM ATMOSPHERIC AIR INTO ALVEOLI

- Partial pressure of oxygen in the atmospheric air is ***159 mm Hg*** whereas in the alveoli, it is ***104 mm Hg***.
- Due to the ***pressure gradient of 55 mm Hg***, oxygen easily enters into the alveoli from atmospheric air.

FROM ALVEOLI INTO BLOOD

- As the blood passes through the pulmonary capillary, RBC is exposed to oxygen just for 0.75 sec (at rest) & only for 0.25 sec (during intense exercise).
- So, diffusion of oxygen from alveoli into the blood must be quicker & effective which is possible because of pressure gradient.
- Partial pressure of oxygen in the pulmonary capillary is ***40 mm Hg*** & ***104 mm Hg*** in the alveoli.
- ***Pressure gradient of 64 mm Hg*** facilitates diffusion of oxygen from alveoli into the blood.


Alveolus
$_pO_2$ = 104 mm Hg
O_2
Arterial end
Venous end
Capillary
$_pO_2$ = 40 mm Hg
O_2 content = 14 mL%
$_pO_2$ = 104 mm Hg
O_2 content = 19 mL%

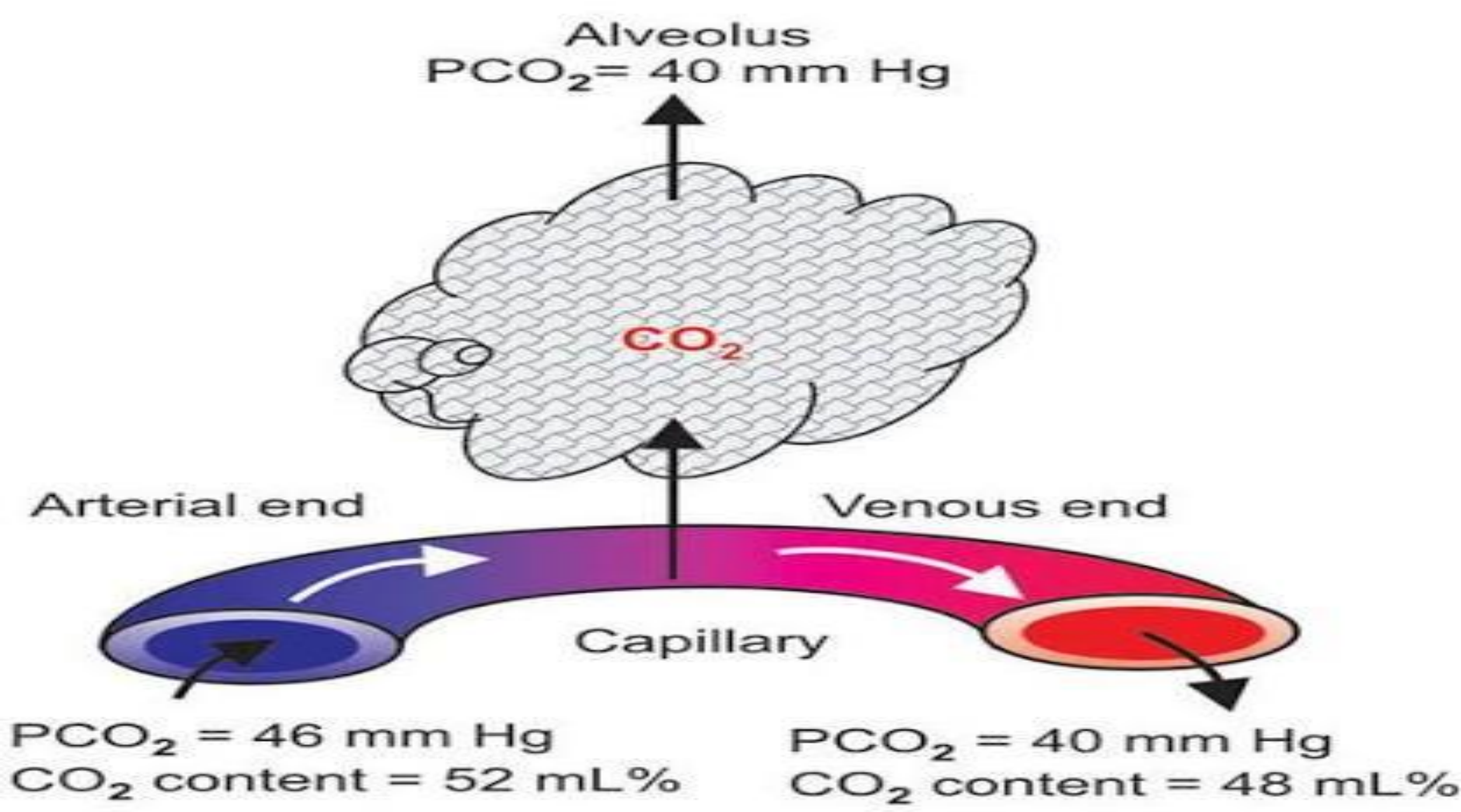
DIFFUSION OF CARBON DIOXIDE

FROM BLOOD INTO ALVEOLI

- Partial pressure of CO_2 in the blood is ***46 mm Hg*** whereas in alveoli, it is ***40 mm Hg***.
- ***Pressure gradient of 6 mm Hg*** causes the diffusion of CO_2 from blood into the alveoli.

FROM ALVEOLI INTO ATMOSPHERIC AIR

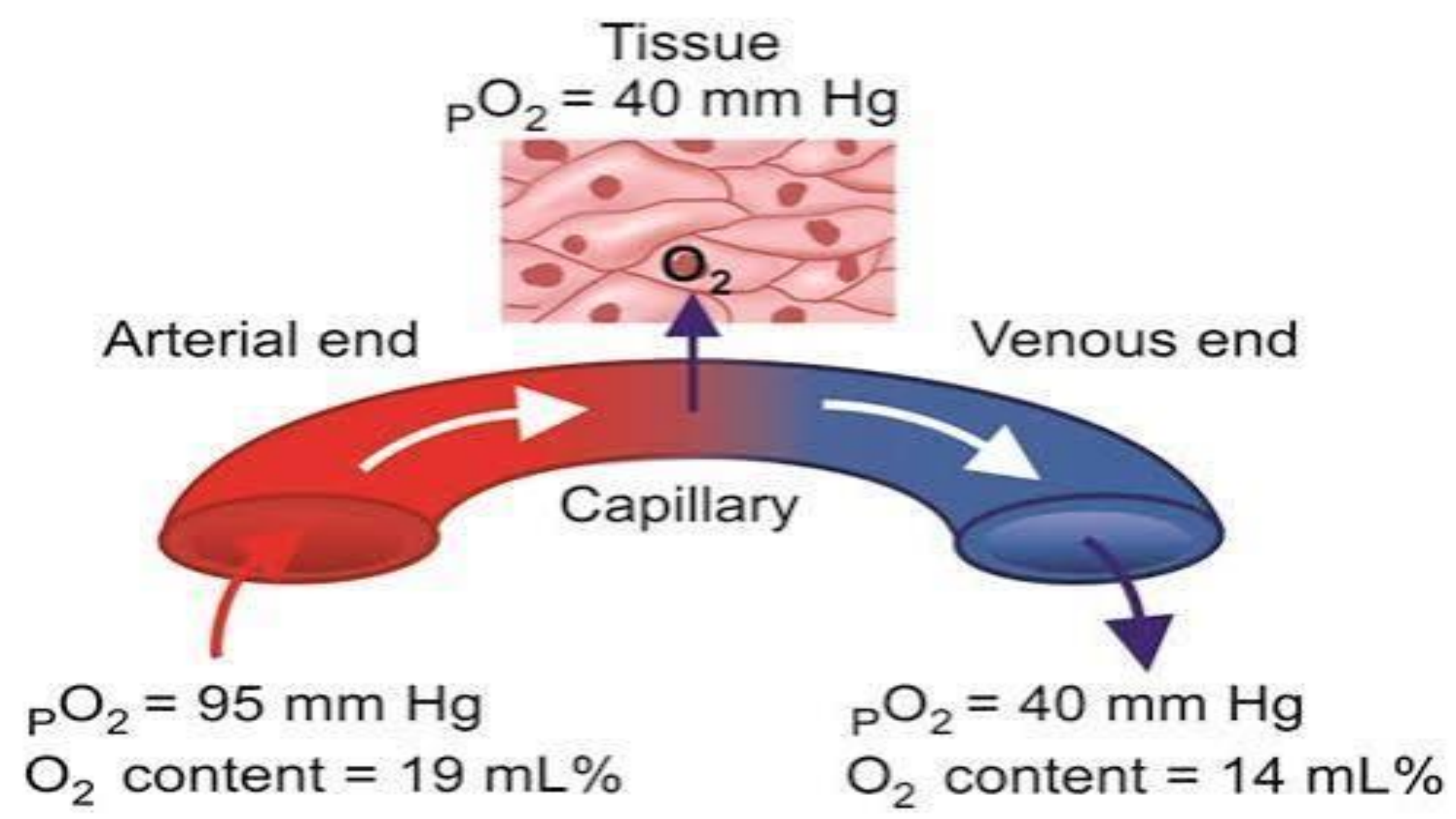
- In alveoli, partial pressure of CO_2 is ***40 mm Hg*** & partial pressure of CO_2 in the atmospheric air is very low i.e. about ***0.3 mm Hg***.
- This huge pressure gradient causes carbon dioxide to diffuse easily from alveoli into the atmospheric air.


Alveolus
PCO_2= 40 mm Hg
CO_2
Arterial end
Venous end
Capillary
PCO_2 = 46 mm Hg
CO_2 content = 52 mL%
PCO_2 = 40 mm Hg
CO_2 content = 48 mL%

EXCHANGE OF RESPIRATORY GASES AT TISSUE LEVEL

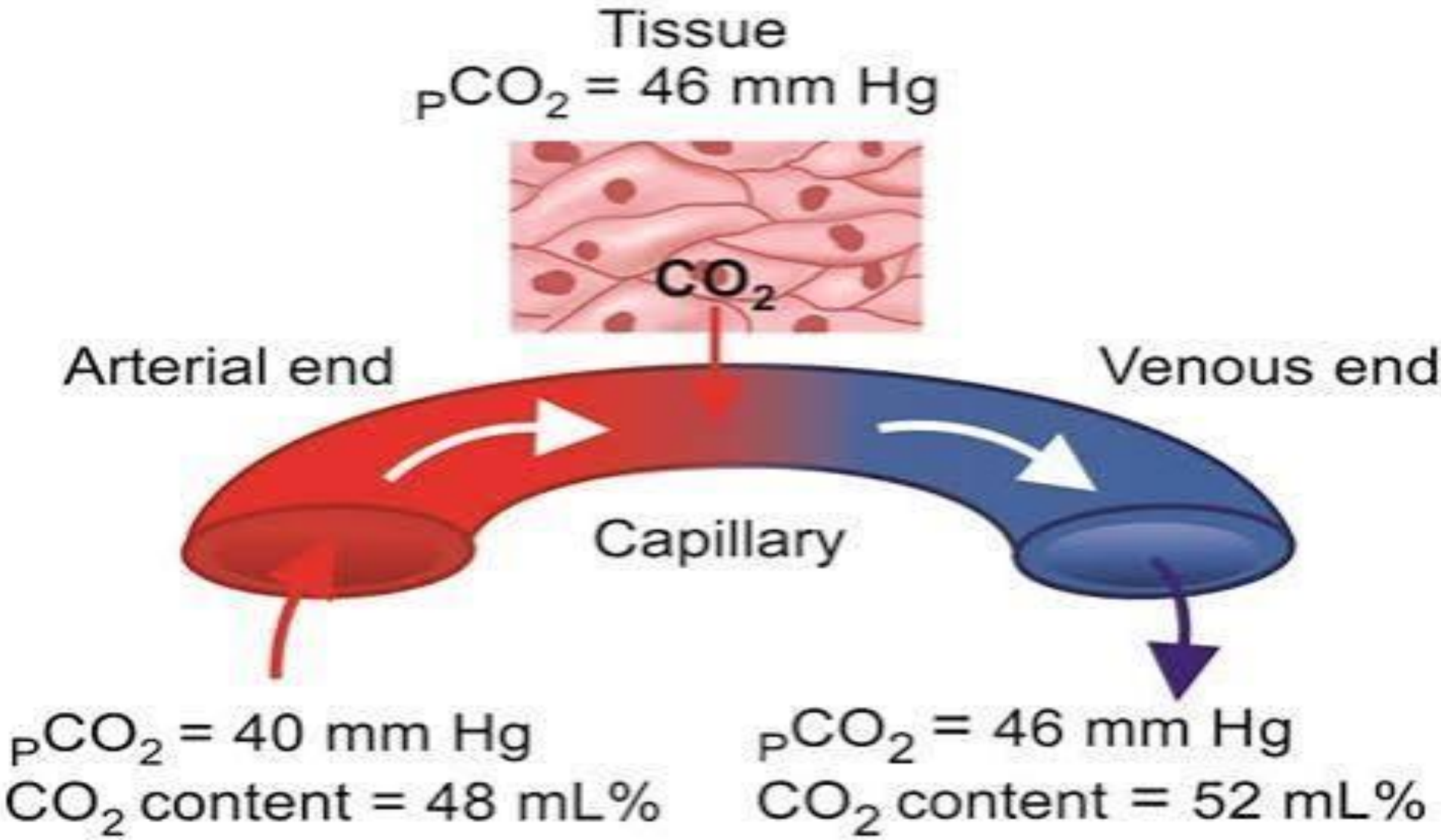
DIFFUSION OF O_2 FROM BLOOD INTO THE TISSUES

- Partial pressure of oxygen at venous end of pulmonary capillary is ***104 mm Hg*** however, at arterial end of systemic capillary it is ***95 mm Hg***.
- This difference is due to *physiological shunt* in the lungs causing venous admixture & 2 % of blood reaches the heart without being oxygenated.
- Partial pressure of oxygen in the tissues is ***40 mm Hg*** due to continuous metabolic activity & constant utilization of oxygen.
- ***Pressure gradient of about 55 mm Hg*** causes diffusion of oxygen into the tissues.
- Diffusion of oxygen from blood to tissues is *5 ml/100 ml* of blood as oxygen content in arterial blood is ***19 ml %*** & in venous blood, it is ***14 ml %.***


Tissue
$_PO_2$ = 40 mm Hg
O_2
Arterial end
Venous end
Capillary
$_PO_2$ = 95 mm Hg
O_2 content = 19 mL%
$_PO_2$ = 40 mm Hg
O_2 content = 14 mL%

DIFFUSION OF CO_2 FROM TISSUES INTO BLOOD

- As a result of continuous metabolic activity, CO_2 is continuously produced in the tissues.
- So, partial pressure of CO_2 in the cells of tissues is high i.e. about ***46 mm Hg*** & partial pressure of CO_2 in arterial blood is low i.e. ***40 mm Hg***.
- ***Pressure gradient of 6 mm Hg*** is responsible for diffusion of carbondioxide from tissues into the blood.
- Carbondioxide content in arterial blood is ***48 ml %*** & in venous blood, it is ***52 ml %*** so, diffusion of carbondioxide from tissues to blood is ***4 ml/100 ml*** of blood.


Tissue
$_PCO_2$ = 46 mm Hg
CO_2
Arterial end
Venous end
Capillary
$_PCO_2$ = 40 mm Hg
CO_2 content = 48 mL%
$_PCO_2$ = 46 mm Hg
CO_2 content = 52 mL%

RESPIRATORY EXCHANGE RATIO (RER)

- It is defined as:

The ratio between net output of carbondioxide from tissues to simultaneous net uptake of oxygen by the tissues.

$$R = \frac{CO_2 \text{ output}}{O_2 \text{ uptake}}$$

- Value of R depends upon the food substance that is metabolized.
- When carbohydrates are utilized, R is 1.0 i.e. amount of carbondioxide produced in the tissues is equal to the amount of oxygen consumed.
- When a person utilizes fats, oxygen reacts with fat & combines with hydrogen ions to form water so carbondioxide output is less than the oxygen consumed.
- If only protein is utilized, R is 0.803 & when a balanced diet is utilized, R is 0.825.

RESPIRATORY QUOTIENT

- It is the molar ratio of carbondioxide production to oxygen consumption.
- For about 1 hr after meals, respiratory quotient is 1.0 because immediately after taking meal only carbohydrates are utilized by the tissues.
- After utilizing all available carbohydrates, body starts utilizing fats so the respiratory quotient becomes 0.7 & when proteins are metabolized, it becomes 0.8.
- In steady conditions, respiratory exchange ratio is equal to respiratory quotient however during exercise, respiratory quotient increases.

The Respiratory Quotient (RQ)

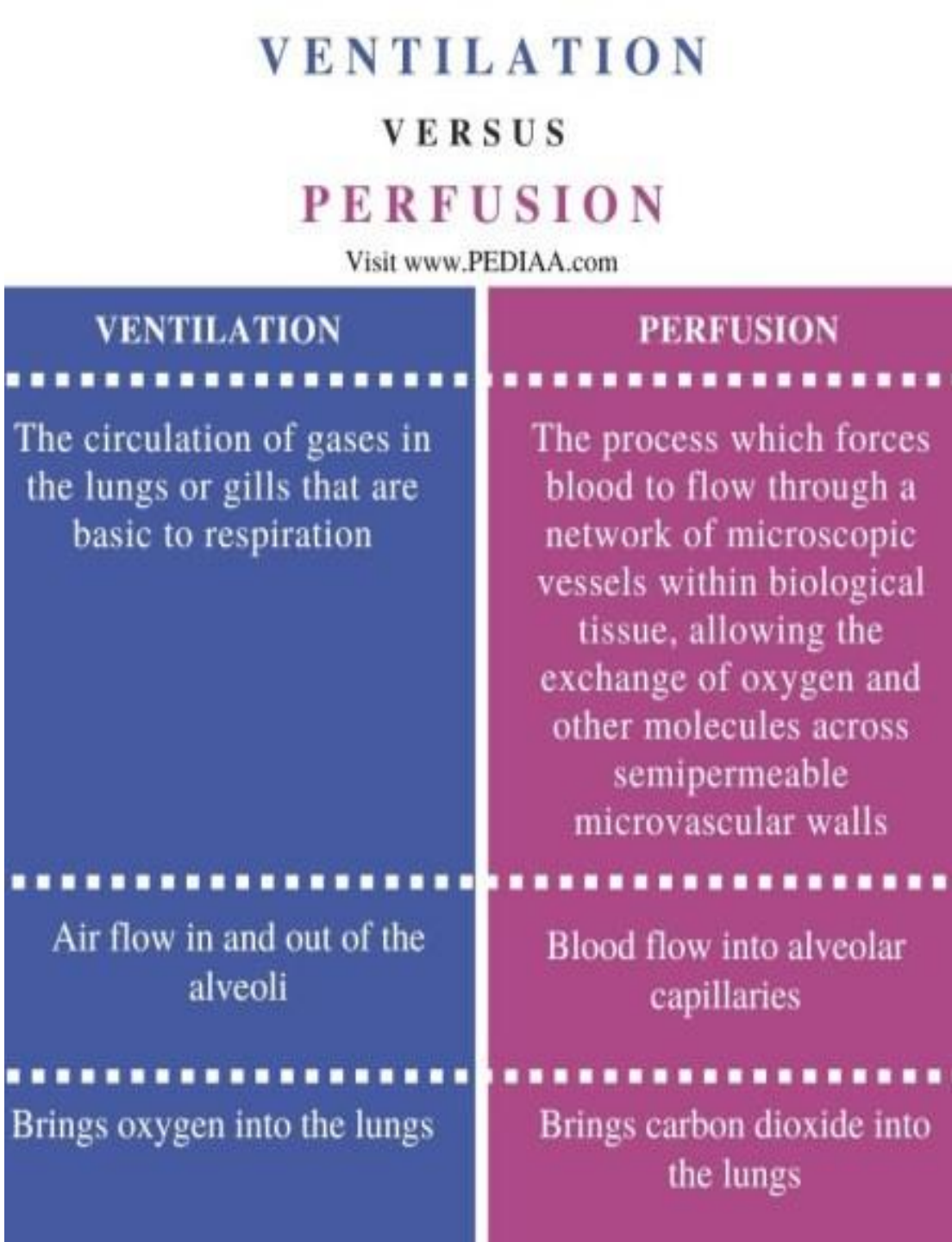
The following table shows the RQ values for different classes of respiratory substrate when they are used for aerobic respiration

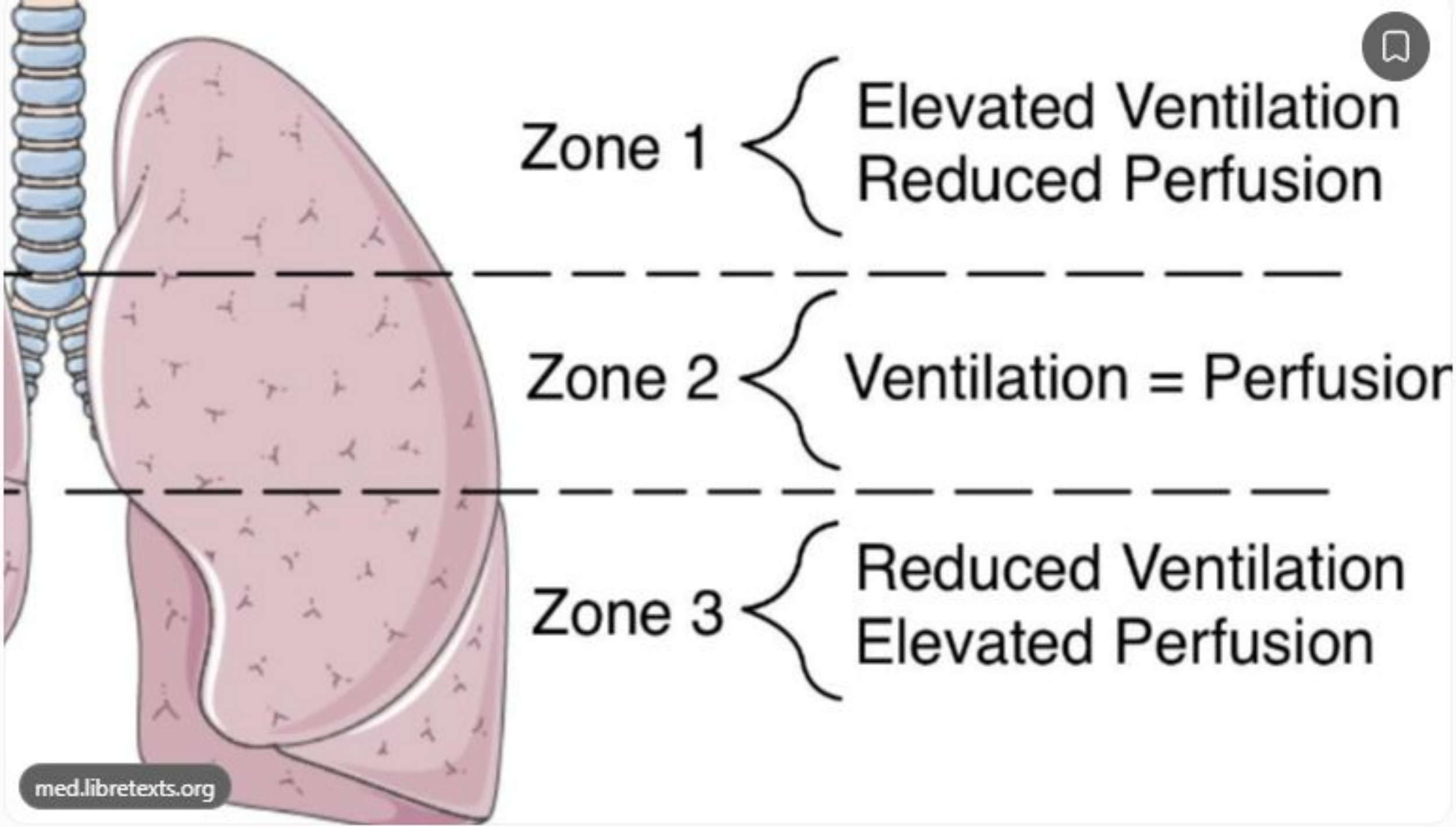
Respiratory Substrate	RQ value
glucose	1.0
fatty acid	0.7
protein	0.9

If any degree of anaerobic respiration occurs RQ values significantly above a value of 1.0 are obtained

VENTILATION - PERFUSION

- Ventilation & perfusion are the two processes that occur in the alveoli of lungs.
- Ventilation is the exchange of air between the lungs & the atmosphere(inhalation & exhalation) whereas perfusion is the process that delivers blood efficiently to the pulmonary capillaries for the gas exchange.
- Ventilation provides oxygen to the alveoli while perfusion brings blood to the pulmonary capillaries for oxygen uptake & carbon dioxide removal.
- Ventilation & perfusion must be appropriately matched for efficient gas exchange across the lungs.




Zone 1
Elevated Ventilation
Reduced Perfusion
Zone 2
Ventilation = Perfusion
Zone 3
Reduced Ventilation
Elevated Perfusion
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Ventilation/Perfusion (V/Q) Ratio

- **It is the ratio of alveolar ventilation and the amount of blood that perfuse the alveoli.**

Calculation:

V/Q= *alveolar ventilation ÷ pulmonary blood flow*

Alveolar ventilation= 4200ml/minute

Pulmonary blood flow=5000ml/minute

So, V/Q= 0.84

VENTILATION-PERFUSION (V/Q) MISMATCH

- V/Q mismatch occurs when ventilation & perfusion are not appropriately matched in different regions of the lungs.
- As a result, some alveoli receive too little air for their blood flow(low V/Q) while others receive more air than their blood flow(high V/Q).
- Both can impair pulmonary gas exchange & contribute to hypoxia.

1. LOW V/Q RATIO (SHUNT LIKE EFFECT)

- Ventilation decreases causing less oxygen entering the alveoli while perfusion is relatively maintained.
- Alveolar oxygen partial pressure falls as a result transfer of oxygen into the blood decreases & the blood leaving the affected region has a lower oxygen content*(hypoxia).*

Causes of low V/Q ratio

- Asthma, chronic obstructive pulmonary disease(COPD), pneumonia, pulmonary edema.

2. HIGH V/Q RATIO (DEAD SPACE VENTILATION)

- Perfusion decreases causing alveolar ventilation to be relatively wasted because less blood is available for gas exchange.
- Alveolar carbon dioxide also decreases & overall gas exchange becomes less efficient.

Causes of high V/Q ratio

- Pulmonary embolism, reduced pulmonary blood flow, emphysema, forms of pulmonary vascular disease.

SHUNT VERSUS DEAD SPACE

SHUNT	DEAD SPACE
The pathological condition, which results when the alveoli of the lungs are perfused with blood as normal, but the ventilation fails to supply the perfused region	The volume of air, which does not take part in the gas exchange as it remains in the conducting airways or reaches alveoli, which are not perfused
Blood Supply: Normal	Blood Supply: Poor
Ventilation: Poor	Ventilation: Normal
Types: Anatomical shunt and capillary shunt	Types: Anatomical dead space and alveolar dead space
Causes: Pneumonia and pulmonary edema, tissue trauma, atelectasis, mucus plugging, pulmonary arteriovenous fistulas, etc.	Causes: Cardiovascular shock, emphysema, pulmonary embolism (PE), etc.

EFFECTS OF VENTILATION/PERFUSION (V/Q) MISMATCH ON GAS EXCHANGE

1. *Hypoxemia (low blood oxygen)*

- This is the primary & most common consequence of V/Q mismatch when the blood passes through poorly ventilated or unventilated alveoli leaves without picking up enough oxygen.

2. *Increased A-a gradient*

- Difference between the partial pressure of oxygen in the alveoli(A) & in the arterial blood(a) increases significantly beyond the normal range(>10-15 mm Hg).

3. *Impact on carbon dioxide*

- As carbon dioxide diffuses much more easily across membranes than oxygen so, carbondioxide levels remain normal or slightly decreased initially because patients can easily compensate by increasing their breathing rate(minute ventilation).
- Hypercapnia (high CO_2) only develops in severe or late stage disease when respiratory muscles tire or dead space is massive.

4. *Increased work of breathing (tachypnea)*

- The body senses lower oxygen or poor ventilation & drives the person to breathe faster & deeper to compensate.

POST ASSESSMENT

Q1. According to fick's law, the rate of gas diffusion is directly proportional to:

a) Membrane thickness
b) Molecular weight of gas
c) Partial pressure gradient
d) Diffusion distance

Q2. Carbon dioxide diffuses across the respiratory membrane more readily than oxygen mainly because CO_2:

a) Has higher molecular weight
b) Binds more strongly to hemoglobin
c) Requires active transport
d) Is more soluble in biological tissues & fluids

CONT..

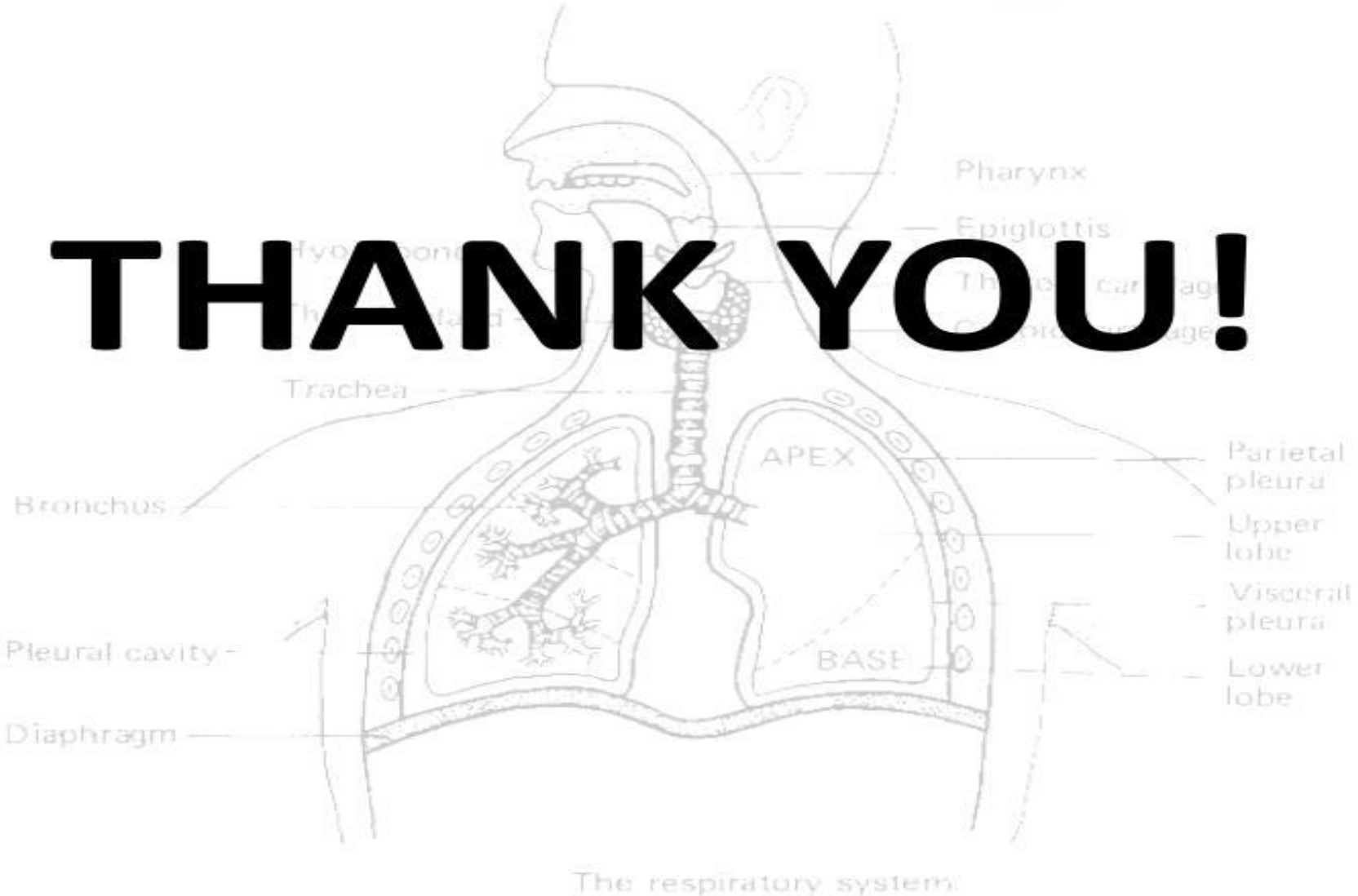
Q3. The normal average ventilation-perfusion ratio for the whole lung at rest is approximately:

a) 0.1
b) 0.3
c) 0.8
d) 1.5

Q4. A low V/Q ratio occurs when:

a) Ventilation is high relative to perfusion
b) Perfusion is absent but ventilation is normal
c) Ventilation is low relative to perfusion
d) Both ventilation & perfusion are absent

THANK YOU!



The respiratory system